



Bluetooth audio for Home

With the latest update, Google's Home smart-home speaker finally enables Bluetooth audio streaming. Find out more: <http://digit.in/GHome2>



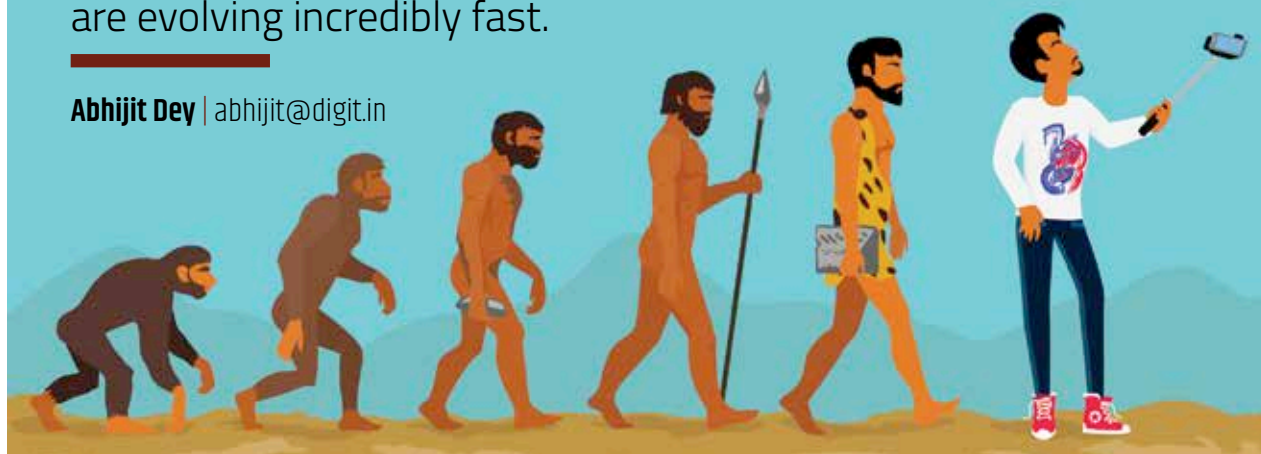
300-mile Hyundai electric car

South Korean auto giant Hyundai is looking to build an electric vehicle by 2021 that will travel 310 miles on a single charge. <http://digit.in/Hyundai300>

Rapid evolution is upon us

Contrary to popular belief, some species are evolving incredibly fast.

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volution has been going on since the first forms of life appeared on the planet billions of

years ago. We have observed the creation and extinction of several species in the past through fossil records and we've seen it even in our lifetime. With evolution, you've probably known that it's an incredibly long process occurring over geological times, spanning thousands and millions of years. As time goes on and the habitat changes, new species are created (speciation) or changes occur within the same species (anagenesis). But researchers have been observing a different phenomenon where species are rapidly evolving under pressure posed by several factors including climate change, commercialisation, pollution, etc. But before we try to understand this new phenomenon, let's first explore what evolution is.

EVOLUTION TAKES A LONG TIME

Although it's insanely difficult to actually measure the average time required for evolution, a study from Oregon

State University suggests that it takes a minimum of one million years. This holds true for long-lasting changes only and not the constantly occurring tiny changes. The reason behind this is because some of those little changes don't necessarily last long. Mechanisms such as mutation, migration, natural and artificial selection, are collectively responsible for evolution. Several traits are developed gradually and the ones that stand the test of time, at least one million years later based on the study, will result in evolution that sticks around. Another thing to note is that even though species might undergo tiny changes in physical traits, there are chances that those traits might not make it to the next generation. For example, if a frog speciates to develop longer legs but doesn't reproduce or gets eaten by a predator, the newly evolved genes responsible for the change won't be passed along to the next generation.

Sometimes, minor genetic changes might range between a few years to something as short as just a few hours. This is dependent on the life cycle of the species. Organisms with high re-

productive rates result in the creation of numerous generations in the span of a few days and months. This is observed in microorganisms and considering their size, it's justified. However, it's being observed that bigger organisms such as lizards, rodents, fish, etc., are evolving quicker than expected, in the span of a few decades. Scientists call it "rapid evolution" and more research is being conducted. While earlier it was accepted that it wasn't possible to observe natural evolution in our lifetime, studies of fossil records and experiments that have been carried out to determine the effects of certain factors to evolution are giving us a lot of new information.

RAPID EVOLUTION ISN'T NEW

Last year, fossil hunters in China discovered the skeleton of a toothless marine reptile, that might have lurked the Earth 247 million years ago. Ichthyosaurs are closely related to marine reptiles and they usually have a long snout, teeth and a tail, including fins. Surprisingly, the newly discovered fossil didn't match those features and was in fact toothless with a short snout